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KOTA (RAJASTHAN)
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PHYSICS

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

Simple Harmonic Motion

English Medium

EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

A. General Equation of SHM

1. A particle is oscillating according to the equation $x = 7\cos 0.5\pi t$, where t is in second. The point moves from the position of equilibrium to maximum displacement in time **[SH0177]**

(A) 4.0 sec (B) 2.0 sec (C) 1.0 sec (D) 0.5 sec

2. A simple harmonic oscillator has an amplitude a and time period T. The time required by it to travel from x = a to x = a / 2 is **[SH0178]**

(A) T/ 6 (B) T / 4 (C) T / 3 (D) T / 2

3. A particle of mass 10 grams is executing simple harmonic motion with an amplitude of 0.5 m and periodic time of $(\pi/5)$ seconds. The maximum value of the force acting on the particle is **[SH0179]**

(A) 25 N (B) 5 N (C) 2.5 N (D) 0.5 N

4. A body is executing S.H.M. When its displacement from the mean position is 4 cm and 5 cm, the corresponding velocity of the body is 10 cm/sec and 8 cm/sec. Then the time period of the body is

(A) 2π sec (B) $\pi/2$ sec (C) π sec (D) $3\pi/2$ sec **[SH0180]**

5. A particle executes simple harmonic motion with an amplitude of 4 cm. At the mean position the velocity of the particle is 10 cm/s. The distance of the particle from the mean position when its speed becomes 5 cm/s is **[SH0181]**

(A) $\sqrt{3}$ cm (B) $\sqrt{5}$ cm (C) $2(\sqrt{3})$ cm (D) $2(\sqrt{5})$ cm

6. A particle executes linear simple harmonic motion with an amplitude of 2 cm. When the particle is at 1 cm from the mean position the magnitude of its velocity is equal to that of its acceleration. Then its time period in seconds is **[SH0182]**

(A) $\frac{1}{2\pi\sqrt{3}}$ (B) $2\pi\sqrt{3}$ (C) $\frac{2\pi}{\sqrt{3}}$ (D) $\frac{\sqrt{3}}{2\pi}$

7. In simple harmonic motion, the ratio of acceleration of the particle to its displacement at any time is a measure of **[SH0183]**

(A) Spring constant (B) Angular frequency

(C) (Angular frequency)^2 (D) Restoring force

8. A student says that he had applied a force $F = -k\sqrt{x}$ on a particle and the particle moved in simple harmonic motion. He refuses to tell whether k is a constant or not. Assume that he has worked only with positive x and no other force acted on the particle. **[SH0184]**

(A) As x increases k increases (B) As x increases k decreases

(C) As x increases k remains constant. (D) The motion cannot be simple harmonic.

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9. The magnitude of average acceleration in half time period from equilibrium position in a simple harmonic motion is **[SH0185]**

(A) $\frac{2A\omega^2}{\pi}$ (B) $\frac{A\omega^2}{2\pi}$ (C) $\frac{A\omega^2}{\sqrt{2}\pi}$ (D) Zero

10. Values of the acceleration $\ddot{x}$ of a particle moving in simple harmonic motion as a function of its displacement x are given in the table below. **[SH0186]**

$\ddot{x}\left(mm/s^2\right)$	16	8	0	–8	–16
x (mm)	–4	–2	0	2	4

The period of the motion is

(A) $\frac{1}{\pi}$s (B) $\frac{2}{\pi}$s (C) $\frac{\pi}{2}$s (D) π s

11. A particle is executing simple harmonic motion given by $x = 5\sin(4t - \pi/6)$. The velocity of the particle when its displacement is 3 units is **[SH0187]**

(A) $\frac{2\pi}{3}$*units* (B) $\frac{5\pi}{6}$*units* (C) 20 units (D) 16 units

12. The motion of a particle is expressed by the equation $a = -bx$, where x is the displacement from the mean position, α is the acceleration and b is a constant. The periodic time is **[SH0188]**

(A) $2\pi/b$ (B) $2\pi/\sqrt{b}$ (C) $2\pi\sqrt{b}$ (D) $2\sqrt{\pi/b}$

13. A body of mass 5 g is executing S.H.M with amplitude 10 cm. Its maximum velocity is 100 cm/s. Its velocity will be 50 cm/s at a displacement from the mean position equal to **[SH0189]**

(A) 5 cm (B) $5\sqrt{3}$ *cm* (C) 10 cm (D) $10\sqrt{3}$ *cm*

14. A simple harmonic oscillator has amplitude A and time period T. Its maximum speed is **[SH0190]**

(A) $\frac{4A}{T}$ (B) $\frac{2A}{T}$ (C) $\frac{4\pi A}{T}$ (D) $\frac{2\pi A}{T}$

15. A simple harmonic oscillator has as period of 0.01s and an amplitude of 0.2 m. The magnitude of the velocity in m/s at the centre of oscillation is **[SH0191]**

(A) 20π (B) 40π (C) 60π (D) 80π

16. A particle is executing S.H.M. Then the graph of acceleration as a function of displacement is

(A) straight line (B) circle (C) ellipse (D) hyperbola **[SH0192]**

17. The displacement equation of an oscillator is $y = 5\sin(0.2\pi t + 0.5\pi)$ in SI units. The time period of oscillation is **[SH0193]**

(A) 10s (B) 1 s (C) 0.2 s (D) 0.5 s

18. A particle executing S.H.M has an acceleration of $64\ cm/s^2$ when its displacement is 4 cm. Its time period, in seconds, is **[SH0194]**

(A) $\pi/2$ (B) $\pi/4$ (C) π (D) 2π

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19. A particle is executing S.H.M of period 4 s. Then the time taken by it to move from the extreme position to half the amplitude is **[SH0195]**

(A) 1/3 s (B) 2/3 s (C) 3/4 s (D) 4/3 s

20. The maximum displacement of a particle executing S.H.M is 1 cm and the maximum acceleration is $(1.57)^2\ cm/s^2$. Its time period is **[SH0196]**

(A) 0.25 s (B) 4.0 s (C) 1.57 s (D) 3.14 s

21. The velocity of a particle, undergoing S.H.M , is v at the mean position. If its amplitude is doubled, the velocity at the mean position will be **[SH0197]**

(A) 2v (B) 3v (C) $2\sqrt{2}$ v (D) 4v

22. The equation of S.H.M of a particle is $\frac{d^2y}{dt^2} + ky = 0$, where k is a positive constant. The time period of motion is given by **[SH0198]**

(A) $\frac{2\pi}{\sqrt{k}}$ (B) $\frac{2\pi}{k}$ (C) $\frac{k}{2\pi}$ (D) $\frac{\sqrt{k}}{2\pi}$

23. Two particles execute S.H.M of the same amplitude and frequency along the same straight line. They pass one another when going in opposite directions each time their displacement is half their amplitude. The phase difference between them is **[SH0199]**

(A) $\pi/3$ (B) $\pi/4$ (C) $2\pi/3$ (D) $3\pi/4$

24. Two particles execute S.H.M of the same time period along the same straight lines. They cross each other at the mean position while going in opposite directions. Their phase difference is **[SH0200]**

(A) $\pi/2$ (B) π (C) $3\pi/2$ (D) 2π

B. Energy of Particle in SHM

25. In a simple harmonic motion **[SH0201]**

(A) the potential energy is always equal to the kinetic energy

(B) the potential energy is never equal to the kinetic energy

(C) the average potential energy in any time interval is equal to the average kinetic energy in that time interval

(D) the average potential energy in one time period is equal to the average kinetic energy in this period.

26. A particle executes simple harmonic motion with a frequency ν. The frequency with which the kinetic energy oscillates is **[SH0202]**

(A) ν/2 (B) ν (C) 2ν (D) zero

27. A point particle of mass 0.1 kg is executing SHM of amplitude 0.1 m. When the particle passes through the mean position, its KE is 8×10^{-3} J. Find the equation of motion of this particle if the initial phase of oscillation is 45°. **[SH0203]**

(A) y = 0.1 cos (3t + π/4) (B) y = 0.1 sin (6t + π/4)

(C) y = 0.1 sin (4t + π/4) (D) y = 0.1 cos (4t + π/4)

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28. A particle of mass 2 kg is executing S.H.M given by $y = 6.0\cos(100t + \pi/4)$ cm. Its maximum kinetic energy is **[SH0204]**

(A) 18 J (B) 36 J (C) $18\times10^4\,J$ (D) $36\times10^4\,J$

29. The total energy of a particle executing S.H.M of amplitude A is proportional to **[SH0205]**

(A) A^2 (B) A^{-2} (C) A (D) 1/A

30. The total energy of a particle executing S.H.M is proportional to the **[SH0206]**

(A) displacement from the equilibrium position

(B) frequency of oscillation

(C) square of the frequency of oscillation

(D) square root of the amplitude of oscillation

31. When a particle executing S.H.M passes through the mean position , it has **[SH0207]**

(A) Maximum K.E and maximum P.E

(B) Maximum K.E and maximum P.E

(C) Maximum K.E and minimum P.E

(D) Minimum K.E and minimum P.E

32. A particle of mass 0.5 kg executes S.H.M. Its energy is 0.04 J. If its time period is π seconds, its amplitude is **[SH0208]**

(A) 10 cm (B) 15 cm (C) 20 cm (D) 40 cm

33. A body executes S.H.M with an amplitude A. Its energy is half kinetic and half potential when the displacement is **[SH0209]**

(A) A/3 (B) A/2 (C) $A/\sqrt{2}$ (D) $A/2\sqrt{2}$

34. For a particle executing S.H.M, the kinetic energy E is given by $E = E_0\cos^2\omega t$. The maximum value of the potential energy is **[SH0210]**

(A) E_0

(B) $E_0\sqrt{2}$

(C) $E_0/2$

(D) undeterminable from the given information

35. A linear harmonic oscillator of force constant 2×10^6 N/m and amplitude 0.01 m has a total mechanical energy of 160 Joule. Its **[SH0211]**

(A) maximum potential energy is 100 J

(B) maximum kinetic energy is 100 J

(C) maximum potential energy is 200 J

(D) minimum potential energy is zero

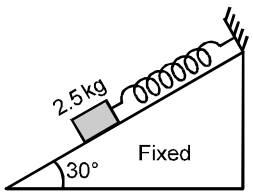
36. The potential energy of a particle executing simple harmonic motion is one-fourth of its maximum value during the oscillation, the displacement of the particle from the equilibrium position in terms of its amplitude a is **[SH0212]**

(A) $a/4$ (B) $a/3$ (C) $a/2$ (D) $2a/3$

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C. Linear SHM

37. A smooth inclined plane having angle of inclination 30° with horizontal has a mass 2.5 kg held by a spring which is fixed at the upper end. If the mass is taken 2.5 cm up along the surface of the inclined plane, the tension in the spring reduces to zero. If the mass is then released, the angular frequency of oscillation in radian per second is **[SH0213]**

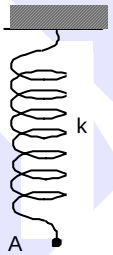


(A) 0.707 (B) 7.07 (C) 1.414 (D) 14.14

38. A ball of mass m kg hangs from a spring of spring constant k. The ball oscillates with a period of T seconds. If the ball is removed, the spring is shortened by **[SH0214]**

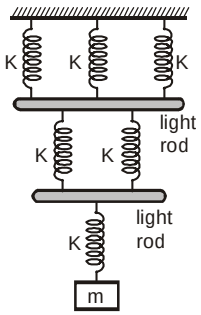
(A) $\frac{gT^2}{(2\pi)^2}$ metre (B) $\frac{3T^2g}{(2\pi)^2}$ metre (C) $\frac{Tm}{k}$ metre (D) $\frac{Tk}{m}$ metre

39. A light spring of stiffness k is hanging from the ceiling. It is stretched by a distance d, then a mass m attached to the end A which is then released. The amplitude of the oscillations will be **[SH0215]**



(A) mg/k (B) 2mg/k (C) d (D) |(mg/k) - d|

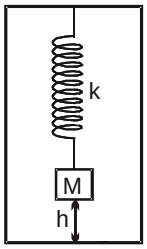
40. The time period of small oscillations of mass m :- **[SH0216]**



(A) $2\pi\sqrt{\frac{m}{6K}}$ (B) $2\pi\sqrt{\frac{11m}{6K}}$ (C) $2\pi\sqrt{\frac{6m}{11K}}$ (D) $2\pi\sqrt{\frac{m}{K}}$

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41. A block is in equilibrium at rest in a lift as shown in figure. Now lift starts accelerating. The minimum magnitude of acceleration of lift for which block will touch the floor of lift, is **[SH0217]**



(A) $\frac{kh}{M}$ (B) $\frac{2kh}{M}$ (C) $\frac{kh}{2M}$ (D) $\frac{3}{2}\frac{kh}{M}$

42. Two springs of force constants k/4 and k/5 are joined in series and then connected to a mass m. The frequency of oscillation of the mass will be **[SH0218]**

(A) $\frac{1}{6\pi}\sqrt{\frac{k}{m}}$ (B) $\frac{3}{2\pi}\sqrt{\frac{k}{m}}$ (C) $\frac{1}{2\pi}\sqrt{\frac{k}{m}}$ (D) $\frac{3}{\pi\sqrt{20}}\sqrt{\frac{k}{m}}$

43. A particle of mass 200g executes S.H.M. The restoring force is provided by a spring of force constant 80 N/m. The time period of oscillation is **[SH0219]**

(A) 0.31 s (B) 0.15 s (C) 0.05 s (D) 0.02 s

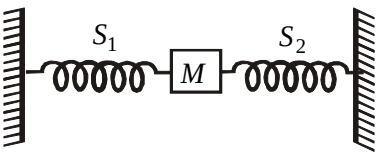
44. The two springs fixed at one end are stretched by 5 cm and 10 cm, respectively, when masses 0.5 kg and 1 kg are suspended at their lower ends. When displaced slightly from their mean positions and released, they will oscillate with time periods in the ratio **[SH0220]**

(A) $1:\sqrt{2}$ (B) 1 : 2 (C) $\sqrt{2}:1$ (D) 2 : 1

45. A mass m attached to a spring oscillates with a period of 3s. If the mass is increased by 1 kg, the period increases by 1s. The initial mass m is **[SH0221]**

(A) $\frac{7}{9}\,kg$ (B) $\frac{9}{7}\,kg$ (C) $\frac{14}{9}\,kg$ (D) $\frac{18}{7}\,kg$

46. In the given diagram, S_1 and S_2 are identical springs. The frequency of oscillation of the mass M is f. If one of the springs is removed, the frequency will be **[SH0222]**



(A) $f/2$ (B) $2f$ (C) $\sqrt{2}f$ (D) $f/\sqrt{2}$

47. The bodies M and N of equal masses are suspended from two separate massless springs of constants k_1 and k_2, respectively. If they oscillate vertically such that their maximum velocities are equal , the ratio of the amplitude of M to that of N is **[SH0223]**

(A) k_1/k_2 (B) k_2/k_1 (C) $\sqrt{k_1/k_2}$ (D) $\sqrt{k_2/k_1}$

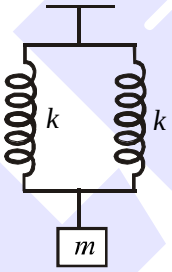
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48. The two spring-mass system, shown in the fig. oscillates with a period T. If only one spring is used, the time period will be **[SH0224]**



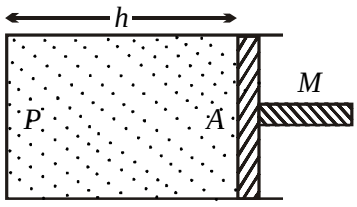
(A) $\frac{T}{\sqrt{2}}$ (B) $\frac{T}{2}$ (C) $\sqrt{2}T$ (D) $2T$

49. The two spring-mass system, shown in fig.oscillates with a period T. If only one spring is used, the time period will be **[SH0225]**



(A) $\frac{T}{\sqrt{2}}$ (B) $\frac{T}{2}$ (C) $\sqrt{2}T$ (D) $2T$

50. A cylindrical piston of mass M and cross-sectional area A slides smoothly inside a long cylinder closed at one end, enclosing a certain mass of a gas. The cylinder is kept with its axis horizontal. If the piston is disturbed from its equilibrium position, it oscillates simple harmonically. The period of oscillation will be. Assume the process to be isothermal. **[SH0226]**



(A) $2\pi\sqrt{\frac{Mh}{PA}}$ (B) $2\pi\sqrt{\frac{MA}{Ph}}$ (C) $2\pi\sqrt{\frac{M}{PAh}}$ (D) $2\pi\sqrt{MPhA}$

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51. One end of a long metallic wire of length L is tied to the ceiling. The other end is tied to a massless spring of spring constant K. A mass m hangs freely from the free end of the spring. The area of cross-section and Young's modulus of the wire are A and Y, respectively. If the mass is slightly pulled down and released, it will oscillate with a time period T equal to **[SH0227]**

(A) $2\pi\sqrt{\frac{m}{K}}$ (B) $2\pi\sqrt{\frac{m(YA+KL)}{YAK}}$ (C) $2\pi\sqrt{\frac{mYA}{KL}}$ (D) $2\pi\sqrt{\frac{mL}{YA}}$

52. A body is on a rough horizontal surface which is moving horizontally in S.H.M of frequency 2 Hz. The coefficient of static friction between the body and the surface is 0.5. The maximum value of the amplitude for which the body will not slip along the surface is approximately **[SH0228]**

(A) 9 cm (B) 6 cm (C) 4.5 cm (D) 3 cm

53. A piece of wood has dimensions a, b and c. Its relative density is d. It is floating in water such that the side a is vertical. If it is pushed down a little and then released, the time period of oscillation will be

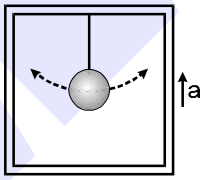
(A) $2\pi\sqrt{\frac{abc}{g}}$ (B) $2\pi\sqrt{\frac{bc}{dg}}$ (C) $2\pi\sqrt{\frac{ad}{g}}$ (D) $2\pi\sqrt{\frac{bcd}{g}}$ **[SH0229]**

54. Springs of constants k, 2k, 4k, 8k,.....2048 k are connected in series. A mass m is attached to one end and the system is allowed to oscillate. The time period is approximately **[SH0230]**

(A) $2\pi\sqrt{\frac{m}{2k}}$ (B) $2\pi\sqrt{\frac{2m}{k}}$ (C) $2\pi\sqrt{\frac{m}{4k}}$ (D) $2\pi\sqrt{\frac{4m}{k}}$

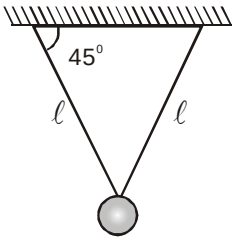
D. Angular SHM

55. A scientist measures the time period of a simple pendulum as T in a lift at rest. If the lift moves up with acceleration as one fourth of the acceleration of gravity, the new time period is **[SH0231]**



(A) $\frac{T}{4}$ (B) $4T$ (C) $\frac{2}{\sqrt{5}}T$ (D) $\frac{\sqrt{5}}{2}T$

56. Time period of small oscillation (in a vertical plane normal to the plane of strings) of the bob in the arrangement shown will be **[SH0232]**



(A) $2\pi\sqrt{\frac{\ell}{g}}$ (B) $2\pi\sqrt{\frac{\ell}{\sqrt{2}g}}$ (C) $2\pi\sqrt{\frac{\sqrt{2}\ell}{g}}$ (D) $2\pi\sqrt{\frac{2\ell}{g}}$

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57. A simple pendulum has time period 2s. The point of suspension is now moved upward according to relation $y = (6t - 3.75\,t^2)$m where t is in second and y is the vertical displacement in upward direction. The new time period of simple pendulum will be **[SH0233]**

(A) 2s (B) 1s (C) 4s (D) None of these

58. Pendulum A is a physical pendulum made from a thin rigid and uniform rod whose length is ℓ. One end of this rod is attached to the ceiling by a frictionless hinge so that rod is free to swing back and forth. Pendulum B is a simple pendulum whose length is also ℓ. The ratio $\frac{T_A}{T_B}$ for small angular oscillations- **[SH0234]**

(A) $\sqrt{\frac{3}{2}}$ (B) $\sqrt{\frac{2}{3}}$ (C) $\frac{2}{3}$ (D) $\frac{3}{2}$

59. The angle made by the string of a simple pendulum with the vertical depends upon time as $\theta = \frac{\pi}{90}\sin \pi t$. Find the length (in m) of the pendulum if $g = \pi^2\,m/s^2$ **[SH0235]**

(A) 1 (B) 2 (C) 3 (D) 4

60. The time period of a simple pendulum at a place where the acceleration due to gravity is g, is T. Its time period at a place where the acceleration due to gravity is 1.02 g, will be **[SH0236]**

(A) T (B) 1.01 T (C) 0.99 T (D) 1.02 T

61. For a simple pendulum, the graph between length and time period will be a **[SH0237]**

(A) hyperbola (B) parabola

(C) straight line (D) none of these

62. A simple pendulum is oscillating in a lift. If the lift starts moving upwards with a uniform acceleration, the period will **[SH0238]**

(A) remain unaffected

(B) be longer

(C) be shorter

(D) may be shorter or longer, depending on the magnitude of acceleration

63. Two simple pendulums having lengths 1 m and 16 m are both given small displacements in the same directions at the same instant. They will again be in phase at the starting position after the shorter pendulum has competed n oscillations where n is **[SH0239]**

(A) 1/4 (B) 4 (C) 5 (D) 16

64. In order to double the period of a simple pendulum **[SH0240]**

(A) Its length should be doubled

(B) Its length should be quadrupled

(C) the mass of its bob should be doubled

(D) the mass of its bob should be quadrupled

65. A body falling freely on a planet covers 8 m in first 2 s. The time period of a one meter long simple pendulum of this planet will be **[SH0241]**

(A) 1.57 s (B) 3.14 s (C) 6.28 s (D) none of these

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66. A simple pendulum suspended from the ceiling of a train has a period T, when the train is at rest. If the train starts moving with a constant acceleration, the time period of the pendulum will **[SH0242]**

(A) increase (B) decrease

(C) remain unaffected (D) become infinite

67. A girl is swinging on a swing in the sitting position. How will the period of swing be affected if she stands up ? **[SH0243]**

(A) The period will now be shorter

(B) The period will now be longer

(C) The period will remain unchanged

(D) The period may be come longer or shorter depending upon the height of the girl

E. Super Position of SHM

68. A particle is subjected to two simple harmonic motions along x and y directions according to x = 3 sin 100 πt, y = 4 sin 100 πt. **[SH0244]**

(A) Motion of particle will be on ellipse traversing it in clockwise direction

(B) Motion of particle will be on a straight line with slope 4/3

(C) Motion will be a simple harmonic motion along x-axis with amplitude 5

(D) Phase difference between two motions is π/2

69. The resultant of two simple harmonic motions $y_1 = 3\sin 314t$ and $y_2 = 4\sin 314t$ has the amplitude

(A) 3 units (B) 4 units (C) 5 units (D) 7 units **[SH0245]**

70. The displacement of a particle in S.H.M. is given by $y = A\sin\omega t + B\cos\omega t$. The amplitude of oscillation is **[SH0246]**

(A) $\frac{A+B}{2}$ (B) $\sqrt{AB}$ (C) $\sqrt{A^2+B^2}$ (D) $\sqrt{\frac{A^2+B^2}{2}}$

71. Two linear simple harmonic motions of equal amplitudes and frequencies ω and 2ω are impressed on a particle along x and y axes respectively. If the initial phase difference between them is $\pi/2$, the resultant path followed by the particle is given by **[SH0247]**

(A) $y^2 = x^2\left(1-\frac{x^2}{A^2}\right)$ (B) $y^2 = 2x^2\left(1-\frac{x^2}{A^2}\right)$ (C) $y^2 = 4x^2\left(1-\frac{x^2}{A^2}\right)$ (D) $y^2 = 8x^2\left(1-\frac{x^2}{A^2}\right)$

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EXERCISE (O-2)

SINGLE CORRECT TYPE QUESTIONS

A. General Equation of SHM

1. The equation of motion of a particle is $x = a\cos(\alpha t)^2$. The motion is

(A) periodic but not oscillatory. (B) periodic and oscillatory.

(C) oscillatory but not periodic. (D) neither periodic nor oscillatory.

SH0037

2. **Statement 1** : Position–time equation of a particle moving along x–axis is $x=4+6\sin\omega t$. Under this situation, motion of particle is not simple harmonic.

and

Statement 2 : $\frac{d^2x}{dt^2}$ for the given equation is not proportional to –x.

(A) Statement–1 is True, Statement–2 is True ; Statement–2 is a correct explanation for Statement–1.

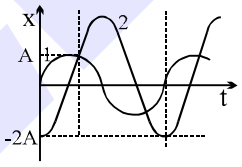
(B) Statement–1 is True, Statement–2 is True ; Statement–2 is not a correct explanation for Statement–1.

(C) Statement–1 is True, Statement–2 is False.

(D) Statement–1 is False, Statement–2 is True.

SH0038

3. The oscillations represented by curve 1 in the graph are expressed by equation $x = A\sin\omega t$. The equation for the oscillations represented by curve 2 is expressed as:



(A) $x = 2A\sin(\omega t - \pi/2)$ (B) $x = 2A\sin(\omega t + \pi/2)$

(C) $x = -2A\sin(\omega t - \pi/2)$ (D) $x = A\sin(\omega t - \pi/2)$

SH0040

4. The time taken by a particle performing SHM to pass from point A to B where its velocities are same is 2 seconds. After another 2 seconds it returns to B. The time period of oscillation is (in seconds)

(A) 2 (B) 8 (C) 6 (D) 4

SH0045

5. A particle performing SHM is found at its equilibrium at t = 1sec. and it is found to have a speed of 0.25 m/s at t = 2 sec. If the period of oscillation is 6 sec. Calculate amplitude of oscillation

(A) $\frac{3}{2\pi}$ m (B) $\frac{3}{4\pi}$ m (C) $\frac{6}{\pi}$ m (D) $\frac{3}{8\pi}$

SH0047

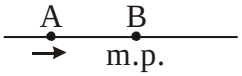
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6. Two particles execute SHM of same amplitude of 20 cm with same period along the same line about the same equilibrium position. The maximum distance between the two is 20 cm. Their phase difference in radians is :-

(A) $\frac{2\pi}{3}$ (B) $\frac{\pi}{2}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{4}$

SH0049

7. A particle is oscillating simple harmonically with angular frequency ω and amplitude A . It is at a point (A) at certain instant (shown in figure). At this instant it is moving towards mean position (B). It takes time t to reach mean position (B). If time period of oscillation is T, the average speed between A and B is :-



(A) $\frac{A\sin\omega t}{t}$ (B) $\frac{A\cos\omega t}{t}$ (C) $\frac{A\sin\omega t}{T}$ (D) $\frac{A\cos\omega t}{T}$

SH0052

8. Speed v of a particle moving along a straight line, when it is at a distance x from a fixed point on the line is given by $v^2 = 108 - 9x^2$ (all quantities in S.I. unit). Then

(A) The motion is uniformly accelerated along the straight line

(B) The magnitude of the acceleration at a distance 3 cm from the fixed point is 0.27 m/s^2.

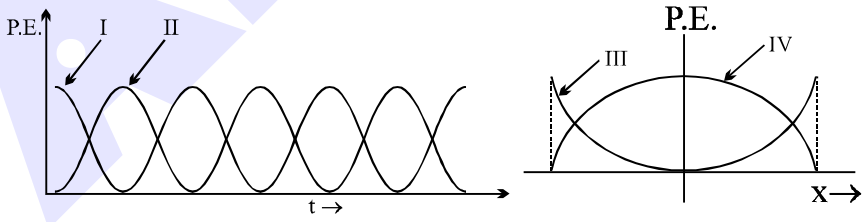
(C) The motion is simple harmonic about x = $\sqrt{12}$ m.

(D) The maximum displacement from the fixed point is 4 cm.

SH0054

B. Energy of Particle in SHM

9. A particle is executing SHM according to $x = a\cos\omega t$. Then which of the graphs represents variations of potential energy : **[IIT JEE (Scr) 2003]**



(A) (I) & (III) (B) (II) & (IV) (C) (I) & (IV) (D) (II) & (III)

SH0055

10. If the potential energy of a harmonic oscillator of mass 2 kg on its equilibrium position is 5 joules and the total energy is 9 joules. If the amplitude is one meter then period of the oscillator (in sec) is:

(A) 1.5 (B) 3.14 (C) 6.28 (D) 4.67

SH0057

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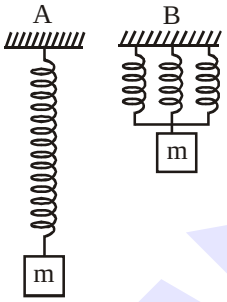
11. Kinetic energy of a particle executing simple harmonic motion in straight line is pv^2 and potential energy is qx^2, where v is speed at distance x from the mean position. It time period is given by the expression :-

(A) $2\pi\sqrt{\frac{q}{p}}$ (B) $2\pi\sqrt{\frac{p}{q}}$ (C) $2\pi\sqrt{\frac{q}{p+q}}$ (D) $2\pi\sqrt{\frac{p}{p+q}}$

SH0058

C. Linear SHM

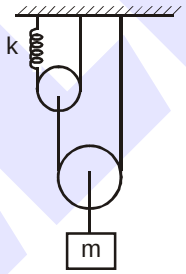
12. The springs in figure A and B are identical but length of spring in A is three times than the length of each spring in B. The ratio of period T_A/T_B is :-



(A) $\sqrt{3}$ (B) 1/3 (C) 3 (D) $1/\sqrt{3}$

SH0060

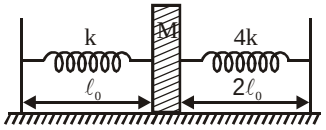
13. What is the period of small oscillations of the block of mass m if the springs are ideal and pulleys are massless ?



(A) $\frac{\pi}{2}\sqrt{\frac{m}{k}}$ (B) $\frac{\pi}{2}\sqrt{\frac{m}{2k}}$ (C) $\frac{\pi}{2}\sqrt{\frac{2m}{k}}$ (D) $\pi\sqrt{\frac{m}{2k}}$

SH0065

14. A block of mass M is kept in gravity free space and touches the two springs as shown in the figure. Initially springs are in their natural lengths. Now, the block is shifted ($\ell_0/2$) from the given position in such a way that it compresses a spring and is released. The time-period of oscillation of mass will be:-



(A) $\frac{\pi}{2}\sqrt{\frac{M}{K}}$ (B) $2\pi\sqrt{\frac{m}{5K}}$ (C) $\frac{3\pi}{2}\sqrt{\frac{M}{K}}$ (D) $\pi\sqrt{\frac{M}{2K}}$

SH0066

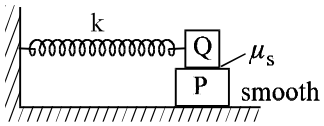
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15. A plank with a small block on top of it is under going vertical SHM. Its period is 2 sec. The minimum amplitude at which the block will separate from plank is :

(A) $\frac{10}{\pi^2}$ (B) $\frac{\pi^2}{10}$ (C) $\frac{20}{\pi^2}$ (D) $\frac{\pi}{10}$

SH0072

16. A block P of mass m is placed on a frictionless horizontal surface. Another block *Q* of same mass is kept on P and connected to the wall with the help of a spring of spring constant k as shown in the figure. μ_s is the coefficient of friction between *P* and *Q*. The blocks move together performing SHM of amplitude *A*. The maximum value of the friction force between *P* and *Q* is :- **[IIT JEE 2004]**



(A) kA (B) $\frac{kA}{2}$ (C) zero (D) $\mu_s mg$

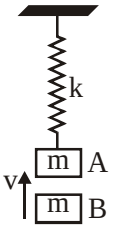
SH0073

17. Vertical displacement of a plank with a body of mass 'm' on it is varying according to law $y = \sin \omega t + \sqrt{3} \cos \omega t$. The minimum value of ω for which the mass just breaks off the plank and the moment it occurs first after t = 0 are given by: (y is positive vertically upwards)

(A) $\sqrt{\frac{g}{2}}, \frac{\sqrt{2}}{6}\frac{\pi}{\sqrt{g}}$ (B) $\frac{g}{\sqrt{2}}, \frac{2}{3}\sqrt{\frac{\pi}{g}}$ (C) $\sqrt{\frac{g}{2}}, \frac{\pi}{3}\sqrt{\frac{2}{g}}$ (D) $\sqrt{2g}, \sqrt{\frac{2\pi}{3g}}$

SH0076

18. Block A is hanging from a vertical spring and is at rest. Block B strikes the block A with velocity v and sticks to it. Then the value of v for which the spring just attains natural length is

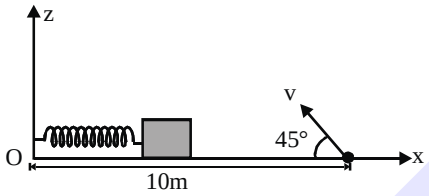


(A) $\sqrt{\frac{60mg^2}{k}}$ (B) $\sqrt{\frac{6mg^2}{k}}$ (C) $\sqrt{\frac{10mg^2}{k}}$ (D) $\sqrt{\frac{8mg^2}{k}}$

SH0077

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19. A small block is connected to one end of a massless spring of un-stretched length 4.9 m. The other end of the spring (see the figure) is fixed. They system lies on a horizontal frictionless surface. The block is stretched by 0.2 m and released from rest at t = 0. It then executes simple harmonic motion with angular frequency $\omega = \frac{\pi}{3}$ rad/s. Simultaneously at t = 0, a small pebble is projected with speed v from point P at an angle of 45° as shown in the figure. Point P is at a horizontal distance of 10m from O. If the pebble hits the block at t = 1s, the value of v is:- (take g = 10 m/s^2) **[IIT-JEE 2012]**



(A) $\sqrt{50}$ m/s (B) $\sqrt{51}$ m/s (C) $\sqrt{52}$ m/s (D) $\sqrt{53}$ m/s

SH0128

D. Angular SHM

20. A bob is attached to a long, light string. The string is deflected by 3° initially with respect to vertical. The length of the string is 1 m . The value of θ at any time t after the bob released can be approximately written as (Use : g = π^2)

(A) 3° cos πt (B) 3° sin πt (C) 3° sin ($\pi t + \frac{\pi}{6}$) (D) 3° cos ($\pi t + \frac{\pi}{6}$)

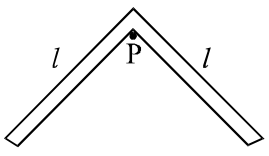
SH0046

21. A rod whose ends are A & B of length 25 cm is hanged in vertical plane. When hanged from point A and point B the time periods calculated are 3 sec & 4 sec respectively. Given the moment of inertia of rod about axis perpendicular to the rod is in ratio 9 : 4 at points A and B. Find the distance of the centre of mass from point A.

(A) 9 cm (B) 5 cm (C) 25 cm (D) 20 cm

SH0061

22. A system of two identical rods (L-shaped) of mass *m* and length *l* are resting on a peg P as shown in the figure. If the system is displaced in its plane by a small angle θ, find the period of oscillations:



(A) $2\pi\sqrt{\frac{\sqrt{2}l}{3g}}$ (B) $2\pi\sqrt{\frac{2\sqrt{2}l}{3g}}$ (C) $2\pi\sqrt{\frac{2l}{3g}}$ (D) $3\pi\sqrt{\frac{l}{3g}}$

SH0062

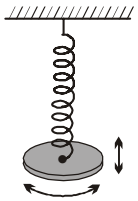
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23. A ring of diameter 2m oscillates as a compound pendulum about a horizontal axis passing through a point at its rim. It oscillates such that its centre move in a plane which is perpendicular to the plane of the ring. The equivalent length of the simple pendulum is :-

(A) 2m (B) 4m (C) 1.5m (D) 3m

SH0063

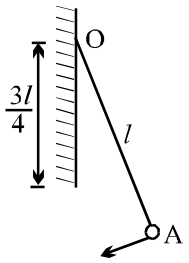
24. A solid disk of radius R is suspended from a spring of linear spring constant k and torsional constant c, as shown in figure. In terms of k and c, what value of R will give the same period for the vertical and torsional oscillations of this system?



(A) $\sqrt{\frac{2c}{k}}$ (B) $\sqrt{\frac{c}{2k}}$ (C) $2\sqrt{\frac{c}{k}}$ (D) $\frac{1}{2}\sqrt{\frac{c}{k}}$

SH0067

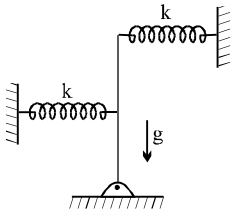
25. A small bob attached to a light inextensible thread of length l has a periodic time T when allowed to vibrate as a simple pendulum. The thread is now suspended from a fixed end O of a vertical rigid rod of length $\frac{3l}{4}$ (as in figure). If now the pendulum performs periodic oscillations in this arrangement, the periodic time will be



(A) $\frac{3T}{4}$ (B) $\frac{T}{2}$ (C) T (D) 2T

SH0075

26. In the figure shown, the spring are connected to the rod at one end and at the midpoint. The rod is hinged at its lower end. Rotational SHM of the rod (Mass m, length L) will occur only if

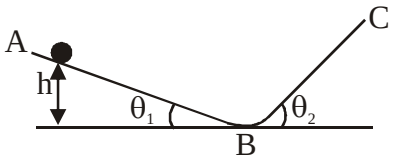


(A) k > mg/3L (B) k > 2mg/3L (C) k > 2mg/5L (D) k > 0

SH0079

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27. A small glass bead of mass m initially at rest starts from a point at height h above the horizontal and rolls down the inclined plane AB as shown. Then it rises along the inclined plane BC. Assuming no loss of energy and sufficient friction for pure rolling, the time period of oscillation of the glass bead is:

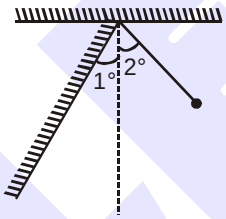


(A) $\sqrt{\frac{8h}{g}}(\sin\theta_1+\sin\theta_2)$ (B) $2\sqrt{\frac{14h}{5g}}\left(\frac{1}{\sin\theta_1}+\frac{1}{\sin\theta_2}\right)$

(C) $\sqrt{\frac{8h}{g}}\left(\frac{1}{\sin\theta_1}+\frac{1}{\sin\theta_2}\right)$ (D) $\sqrt{\frac{8h}{5g}}\left(\frac{1}{\sin\theta_1}+\frac{1}{\sin\theta_2}\right)$

SH0080

28. A simple pendulum of length 1m is allowed to oscillate with amplitude 2°. It collides elastically with a wall inclined at 1° to the vertical. Its time period will be : (use $g = \pi^2$) **SH0248**



(A) 2/3 sec (B) 4/3 sec (C) 2 sec (D) None of these

E. Super Position of SHM

29. The amplitude of the vibrating particle due to superposition of two SHMs,

$y_1 = \sin\left(\omega t + \frac{\pi}{3}\right)$ and $y_2 = \sin\omega t$ is :

(A) 1 (B) $\sqrt{2}$ (C) $\sqrt{3}$ (D) 2

SH0068

30. Two simple harmonic motions $y_1 = A\sin\omega t$ and $y_2 = A\cos\omega t$ are superimposed on a particle of mass m. The total mechanical energy of the particle is:

(A) $\frac{1}{2}m\omega^2A^2$ (B) $m\omega^2A^2$ (C) $\frac{1}{4}m\omega^2A^2$ (D) zero

SH0069

31. Equations $y = 2A\cos^2\omega t$ and $y = A(\sin\omega t + \sqrt{3}\cos\omega t)$ represent the motion of two particles.

(A) Only one of these is S.H.M. (B) Ratio of maximum speeds is 2 : 1

(C) Ratio of maximum speeds is 1 : 1 (D) Ratio of maximum accelerations is 1 : 4

SH0071

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32. A point mass is subjected to two simultaneous sinusoidal displacements in x-direction, $x_1(t) = A\sin\omega t$ and $x_2(t) = A\sin\left(\omega t + \frac{2\pi}{3}\right)$. Adding a third sinusoidal displacement $x_3(t) = B\sin(\omega t + \phi)$ brings the mass to a complete rest. The values of B and ϕ are :- **[IIT-JEE 2011]**

(A) $\sqrt{2}A, \frac{3\pi}{4}$ (B) $A, \frac{4\pi}{3}$ (C) $\sqrt{3}A, \frac{5\pi}{6}$ (D) $A, \frac{\pi}{3}$

SH0125

MULTIPLE CORRECT TYPE QUESTIONS

A. General Equation of SHM

33. For a SHM with given angular frequency, two arbitrary initial conditions are necessary and sufficient to determine the motion completely. These initial conditions may be-

(A) Amplitude and initial phase

(B) Amplitude and total energy of oscillation

(C) Initial phase and total energy of oscillation

(D) Initial position and initial velocity

SH0097

34. A particle is executing SHM between points $-X_m$ and X_m, as shown in figure-I. The velocity V(t) of the particle is partially graphed and shown in figure-II. Two points A and B corresponding to time t_1 and time t_2 respectively are marked on the V(t) curve.

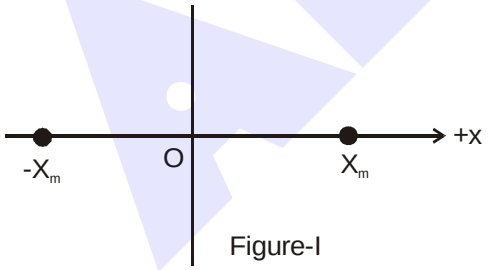


Figure-I

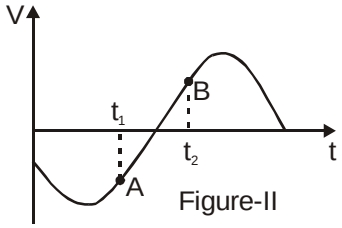


Figure-II

(A) At time t_1 , it is going towards X_m.

(B) At time t_1, its speed is decreasing.

(C) At time t_2, its position lies in between $-X_m$ and O.

(D) The phase difference $\Delta\phi$ between points A and B must be expressed as $90° < \Delta\phi < 180°$.

SH0098

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35. The displacement (in m) of a particle of mass 100 g from its equilibrium position is given by the equation: **SH0249**

$$y = 0.05 \sin 3\pi (5t + 0.4)$$

(A) the time period of motion is 1/30 sec

(B) the time period of motion is 1/7.5 sec

(C) the maximum acceleration of the particle is $11.25\pi^2$ m/s^2

(D) the force acting on the particle is zero when the displacement is 0.05 m.

C. Linear SHM

36. A mass of 0.2kg is attached to the lower end of a massless spring of force-constant 200 N/m, the upper end of which is fixed to a rigid support. Which of the following statements is/are true?

(A) In equilibrium, the spring will be stretched by 1cm.

(B) If the mass is raised till the spring is unstretched state and then released, it will go down by 2cm before moving upwards.

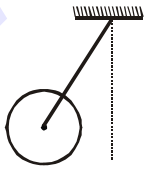
(C) The frequency of oscillation will be nearly 5 Hz.

(D) If the system is taken to the moon, the frequency of oscillation will be the same as on the earth.

SH0105

D. Angular SHM

37. A metal rod of length 'L' and mass 'm' is pivoted at one end. A thin disk of mass 'M' and radius 'R' (< L) is attached at its center to the free end of the rod. Consider two ways the disc is attached: (case A). The disc is not free to rotate about its center and (case B) the disc is free to rotated about its center. The rod-disc system performs SHM in vertical plane after being released from the same displaced position. Which of the following statement(s) is(are) true? **[IIT-JEE 2011]**



(A) Restoring torque in case A = Restoring torque in case B

(B) Restoring torque in case A < Restoring torque in case B

(C) Angular frequency for case A> Angular frequency for case B

(D) Angular frequency for case A< Angular frequency for case B

SH0126

38. The motion of a torsional pendulum is **SH0250**

(A) periodic (B) oscillatory

(C) simple harmonic (D) angular simple harmonic

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E. Super Position of SHM

39. The position vector of a particle that is moving in space is given by

$\vec{r} = (1 + 2\cos 2\omega t)\hat{i} + (3\sin^2 \omega t)\hat{j} + (3)\hat{k}$

in the ground frame. All units are in SI. Choose the correct statement (s) :

(A) The particle executes SHM in the ground frame about the mean position $\left(1, \frac{3}{2}, 3\right)$

(B) The particle executes SHM in a frame moving along the z-axis with a velocity of 3 m/s.

(C) The amplitude of the SHM of the particle is $\frac{5}{2}$ m.

(D) The direction of the SHM of the particle is given by the vector $\left(\frac{4}{5}\hat{i} - \frac{3}{5}\hat{j}\right)$

SH0101

40. A particle moves in x-y plane according to equation $\vec{S} = \left(2\hat{i} + \hat{j}\right)\cos \omega t$. The motion of particle is:-

(A) On straight line (B) On ellipse (C) Periodic (D) SHM

SH0106

41. A 20 gm particle is subjected to two simple harmonic motions **SH0251**

$x_1 = 2 \sin 10\ t$; $x_2 = 4 \sin \left(10t + \frac{\pi}{3}\right)$, where x_1 & x_2 are in metre & t is in sec.

(A) The displacement of the particle at t = 0 will be $2\sqrt{3}$ m

(B) Maximum speed of the particle will be $20\sqrt{7}$

(C) Magnitude of maximum acceleration of the particle will be $200\sqrt{7}$ m/s^2.

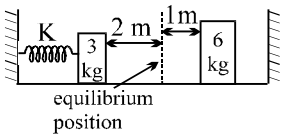
(D) Energy of the resultant motion will be 28 J

COMPREHENSION TYPE QUESTIONS

C. Linear SHM

Paragraph for Question No. 42 and 43

Two blocks of masses 3 kg and 6 kg rest on on a horizontal frictionless surface. The 3 kg block is attached to a spring with a force constant K = 900 N/m which is compressed 2 m initially from its equilibrium position. When 3 kg mass is released, it strikes the 6 kg mass and the two stick together.



42. The common velocity of the blocks after collision is :-

(A) 10 m/s (B) 30 m/s (C) 15 m/s (D) 2 m/s

SH0112

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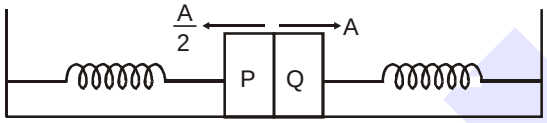
43. The amplitude of resulting oscillation after the collision is :-

(A) $\frac{1}{\sqrt{2}}$ m (B) $\frac{1}{\sqrt{3}}$ m (C) $\sqrt{2}$ m (D) $\sqrt{3}$ m

SH0112

Paragraph for Question no. 44 and 45

Two identical blocks P and Q have mass m each. They are attached to two identical springs initially unstretched. Now the left spring (alongwith P) is compressed by $\frac{A}{2}$ and the right spring (alongwith Q) is compressed by A. Both the blocks are released simultaneously. They collide perfectly inelastically. Initially time period of both the block was T.



44. The time period of oscillation of combined mass is :

(A) $\frac{T}{\sqrt{2}}$ (B) $\sqrt{2}T$ (C) T (D) $\frac{T}{2}$

SH0113

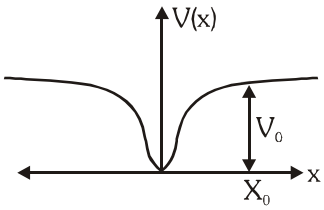
45. The amplitude of combined mass is :

(A) $\frac{A}{4}$ (B) $\frac{A}{2}$ (C) $\frac{2A}{3}$ (D) $\frac{3A}{4}$

SH0113

Paragraph for Question No. 46 to 48

When a particle of mass m moves on the x–axis in a potential of the form $V(x) = kx^2$, it performs simple harmonic motion. The corresponding time period is proportional to $\sqrt{\frac{m}{k}}$, as can be seen easily using dimensional analysis. However, the motion of a particle can be periodic even when its potential energy increases on both sides of $x = 0$ in a way different from kx^2 and its total energy is such that the particle does not escape to infinity. Consider a particle of mass m moving on the x–axis. Its potential energy is $V(x) = \alpha x^4$ ($\alpha > 0$) for $|x|$ near the origin and becomes a constant equal to V_0 for $|x| \geq X_0$ (see figure) **[IIT-JEE 2010]**



46. If the total energy of the particle is E, it will perform periodic motion only if :-

(A) $E < 0$ (B) $E > 0$ (C) $V_0 > E > 0$ (D) $E > V_0$

SH0124

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47. For periodic motion of small amplitude A, the time period T of this particle is proportional to :-

(A) $A\sqrt{\frac{m}{\alpha}}$ (B) $\frac{1}{A}\sqrt{\frac{m}{\alpha}}$ (C) $A\sqrt{\frac{\alpha}{m}}$ (D) $\frac{1}{A}\sqrt{\frac{\alpha}{m}}$

SH0124

48. The acceleration of this particle for $|x| > X_0$ is :-

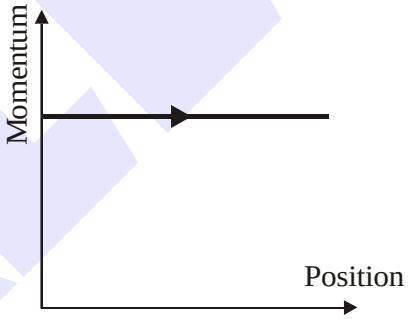
(A) proportional to V_0 (B) proportional to $\frac{V_0}{mX_0}$

(C) proportional to $\sqrt{\frac{V_0}{mX_0}}$ (D) Zero

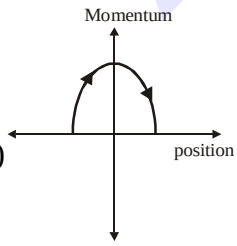
SH0124

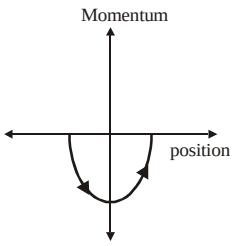
Paragraph for Questions Nos. 49 to 51

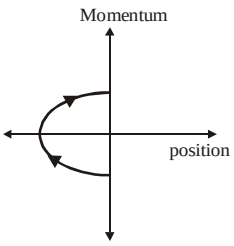
Phase space diagrams are useful tools in analyzing all kinds of dynamical problems. They are especially useful in studying the changes in motion as initial position and momentum are changed. Here we consider some simple dynamical systems in one-dimension. For such systems, phase space is a plane in which position is plotted along horizontal axis and momentum is plotted along vertical axis. The phase space diagram is x(t) vs. p(t) curve in this plane. The arrow on the curve indicates the time flow. For example, the phase space diagram for a particle moving with constant velocity is a straight line as shown in the figure. We use the sign convention in which position or momentum upwards (or to right) is positive and downwards (or to left) is negative. **[IIT-JEE 2011]**

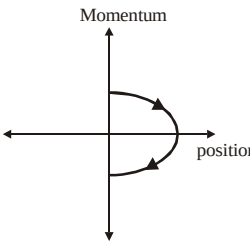


49. The phase space diagram for a ball thrown vertically up from ground is

(A) 

(B) 

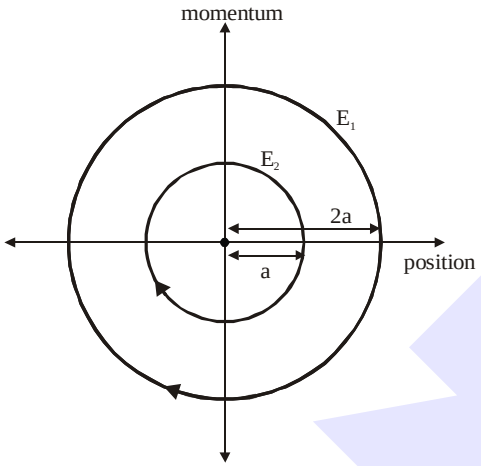
(C) 

(D) 

SH0127

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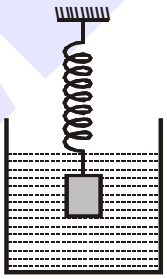
50. The phase space diagram for simple harmonic motion is a circle centered at the origin. In the figure, the two circles represent the same oscillator but for different initial conditions, and E_1 and E_2 are the total mechanical energies respectively. Then

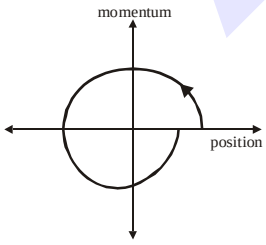


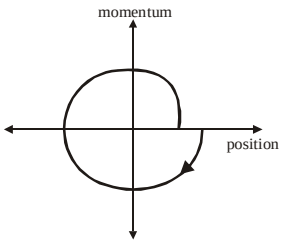
(A) $E_1 = \sqrt{2}E_2$ (B) $E_1 = 2E_2$ (C) $E_1 = 4E_2$ (D) $E_1 = 16E_2$

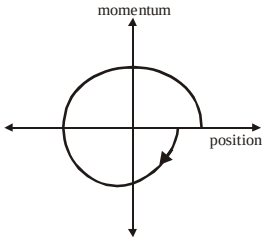
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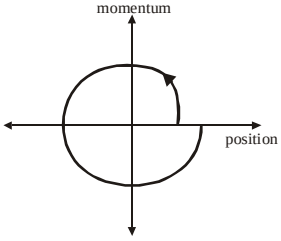
51. Consider the spring-mass system, with the mass submerged in water, as shown in the figure. The phase space diagram for one cycle of this system is



(A) 

(B) 

(C) 

(D) 

SH0127

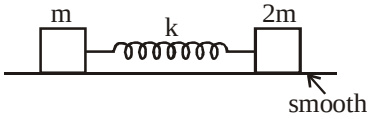
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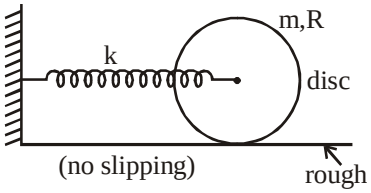
MATRIX MATCH TYPE QUESTION

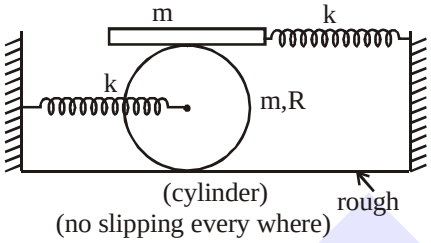
D. Angular SHM

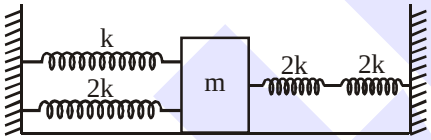
52. In list-I, the systems are performing SHM and in list-II, the time period of SHM is shown then match list-I with list-II.

List-I

(P) 

(Q) 

(R) 

(S) 

List-II

(1) $\left[2\pi\sqrt{\frac{11m}{10k}}\right]$

(2) $2\pi\sqrt{\frac{m}{4k}}$

(3) $2\pi\sqrt{\frac{3m}{2k}}$

(4) $2\pi\sqrt{\frac{2m}{3k}}$

Codes :

	P	Q	R	S
(A)	3	2	1	4
(B)	4	3	2	1
(C)	3	4	2	1
(D)	4	3	1	2

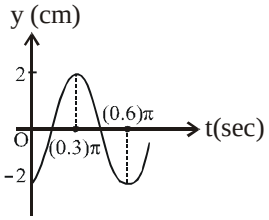
SH0115

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EXERCISE (S)

A. General Equation of SHM

1. Part of a simple harmonic motion is graphed in the figure, where y is the displacement from the mean position. The correct equation describing this S.H.M is :-



SH0001

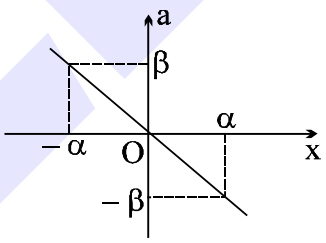
2. The displacement of a body executing SHM is given by $x = A \sin(2\pi t + \pi/3)$. The first time from t = 0 when the velocity is maximum is.

SH0002

3. A body undergoing SHM about the origin has its equation given by $x = 0.2 \cos 5\pi t$. Find its average speed from t = 0 to t = 0.7 sec.

SH0003

4. The acceleration-displacement (a – x) graph of a particle executing simple harmonic motion is shown in the figure. Find the frequency of oscillation.



SH0007

5. A simple harmonic motion has an amplitude A and time period T. Find the time required by it to travel directly from **SH0252**

(a) x = 0 to x = A/2 (b) x = 0 to $x = \frac{A}{\sqrt{2}}$ (c) x = A to x = A/2 (d) $x = -\frac{A}{\sqrt{2}}$ to $x = \frac{A}{\sqrt{2}}$

(e) $x = \frac{A}{\sqrt{2}}$ to x = A

6. A particle executes simple harmonic motion with an amplitude of 10 cm and time period 6 s. At t = 0 it is at position x = 5 cm going towards positive x-direction. Write the equation for the displacement x at time t. Find the magnitude of the acceleration of the particle at t = 4 s. **SH0253**

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B. Energy of Particle in SHM

7. An object of mass 0.2 kg executes SHM along the x-axis with frequency of $(25/\pi)$ Hz. At the point x = 0.04m the object has KE 0.5 J and PE 0.4 J. The amplitude of oscillation is _____.

SH0004

8. A point particle of mass 0.1kg is executing SHM with amplitude of 0.1m. When the particle passes through the mean position, its K.E. is 8×10^{-3}J. Obtain the equation of motion of this particle if the initial phase of oscillation is 45°.

SH0005

9. Potential Energy (U) of a body of unit mass moving in a one-dimension conservative force field is given by $U = (x^2 - 4x + 3)$. All units are in S.I.

(i) Find the equilibrium position of the body.

(ii) Show that oscillations of the body about this equilibrium position is simple harmonic motion & find its time period.

(iii) Find the amplitude of oscillations if speed of the body at equilibrium position is $2\sqrt{6}$ m/s.

SH0006

10. Calculate the ratio between the kinetic energy E_k and the potential energy E_p of an oscillating point for the moment t = T/12, where T = time period. **SH0254**

C. Linear SHM

11. A small body of mass m is fixed to the middle of a stretched string of length 2ℓ. In the equilibrium position the string tension is equal to T_0. Find the angular frequency of small oscillations of the body in the transverse direction. The mass of the string is negligible, the gravitational field is absent. Assume tension in string to be constant.

SH0008

12. A body is in SHM with period T when oscillated from a freely suspended spring. If this spring is cut in two parts of length ratio 1 : 3 & again oscillated from the two parts separately, then the periods are T_1 & T_2 then find T_1/T_2.

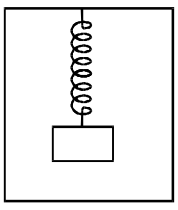
SH0009

13. A mass M attached to a spring, oscillates with a period of 2 sec. If the mass is increased by 2 kg the period increases by one sec. Find the initial mass M assuming that Hooke's Law is obeyed.

SH0012

14. A spring mass system is hanging from the ceiling of an elevator in equilibrium. Elongation of spring is *l*. The elevator suddenly starts accelerating downwards with acceleration g/3, find

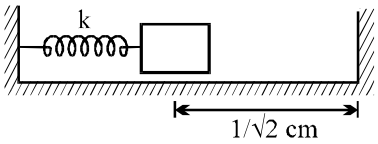
(a) the frequency and (b) the amplitude of the resulting SHM.



SH0013

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15. A block of mass 0.9 kg attached to a spring of force constant k is lying on a frictionless floor. The spring is compressed to $\sqrt{2}$ cm and the block is at a distance $1/\sqrt{2}$ cm from the wall as shown in the figure. When the block is released, it makes elastic collision with the wall and its period of motion is 0.2 sec. Find the approximate value of k.

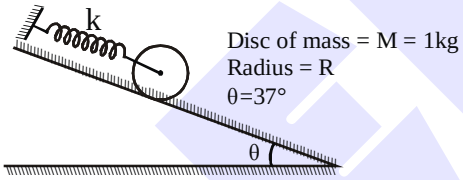


SH0017

16. A body of mass 1 kg is suspended from a weightless spring having force constant 600N/m. Another body of mass 0.5 kg moving vertically upwards hits the suspended body with a velocity of 3.0m/s and get embedded in it. Find the frequency of oscillations and amplitude of motion.

SH0018

17. A disc of mass m is connected to an ideal spring of force constant 'k'. If disc is released from rest, then what is maximum friction force on disc (in N). Assuming friction is sufficient for pure rolling

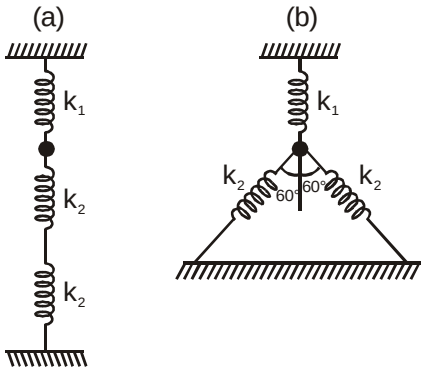


SH0020

18. A particle of mass 5×10^{-5} kg is placed at the lowest point of a smooth parabola having the equation $20x^2 = y$ (x,y in m). Here y is the vertical height. If it is displaced slightly and it moves such that it is constrained to move along the parabola, the angular frequency of oscillation will be, (in rad/s). If your answer is N fill value N/4.

SH0021

19. Mass m is suspended by ideal massless springs in two different ways, indicated by (a) and (b) in the figure. The mass is displaced upwards by a small amount from equilibrium position and is then released resulting in a SHM of the mass in the vertical direction. We denote the oscillation frequencies associated with the two cases (a) and (b) by f_a and f_b respectively. Find $\frac{f_a}{f_b}$. Given $k_2 = 2k_1$.



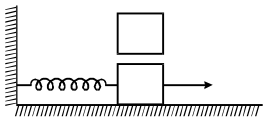
SH0022

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20. A solid sphere of radius R is floating in a liquid of density ρ with half of its volume submerged. If the sphere is slightly pushed and released, it starts performing simple harmonic motion. Find the frequency of these oscillations. **[IIT-JEE 2004]**

SH0023

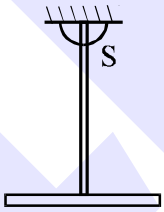
21. A 1 kg block is executing simple harmonic motion of amplitude 0.1 m on a smooth horizontal surface under the restoring force of a spring of spring constant 100 N/m. A block of mass 3 kg is gently placed on it at the instant it passes through the mean position. Assuming that the two blocks move together, find the frequency and the amplitude of the motion. **SH0255**



22. A spring stores 5 J of energy when stretched by 25 cm. It is kept vertical with the lower end fixed. A block fastened to its other end is made to undergo small oscillations. If the block makes 5 oscillations each second, what is the mass of the block ? **SH0256**

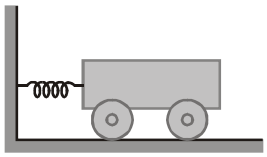
D. Angular SHM

23. Two identical rods each of mass m and length L, are rigidly joined and then suspended in a vertical plane so as to oscillate freely about an axis normal to the plane of paper passing through 'S' (point of suspension). Find the time period of such small oscillations.



SH0010

24. A cart consists of a body and four wheels on frictionless axles. The body has a mass m. The wheels are uniform disks of mass M and radius R. The cart rolls, without slipping, back and forth on a horizontal plane under the influence of a spring attached to one end of the cart (figure). The spring constant is k. Taking into account the moment of inertia of the wheels, find a formula for the frequency of the back and forth motion of the cart.



SH0011

25. (a) Find the time period of oscillations of a torsional pendulum, if the torsional constant of the wire is $K = 10\pi^2$ J/rad. The moment of inertia of rigid body is 10 kg m^2 about the axis of rotation.

(b) A simple pendulum of length l = 0.5 m is hanging from ceiling of a car. The car is kept on a horizontal plane. The car starts accelerating on the horizontal road with acceleration of 5 m/s^2. Find the time period of oscillations of the pendulum for small amplitudes about the mean position.

SH0014

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26. The motion of a simple pendulum is given by

$$\theta = A\cos\left(\sqrt{\frac{g}{\ell}}\,t\right)$$ (symbols have their usual meaning)

(a) Find the tension in the string of this pendulum as a function of time assume that $\theta << \ell$.

(b) At what time is the tension maximum? What is the value of this maximum tension?

SH0015

27. A physical pendulum has the shape of a disk of radius R. The pendulum swings about an axis perpendicular to the plane of the disk and at distance ℓ from the center of the disk.

(a) Show that the frequency of the oscillations of this pendulum is $\omega = \sqrt{\frac{g\ell}{\frac{1}{2}R^2 + \ell^2}}$

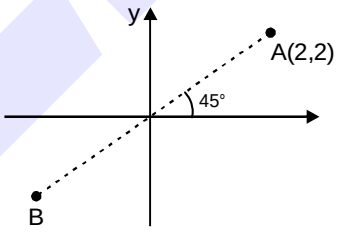
(b) For what value of ℓ is this frequency at a maximum ?

SH0016

28. A uniform disc of radius r is to be suspended through a small hole made in the disc. Find the minimum possible time period of the disc for small oscillations. What should be the distance of the hole from the centre for it to have minimum time period ? **SH0257**

E. Super Position of SHM

29. A particle of mass m = 2 kg executes SHM in xy-plane between points A and B as shown in figure under action of force $\vec{F} = F_x i + F_y j$. Minimum time taken by particle to move from A to B is 1 sec. At t = 0 the particle is at x = 2 and y = 2. Then find the value of F_x as the function of time. **SH0258**



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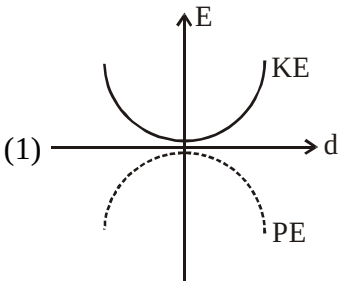
EXERCISE (JM)

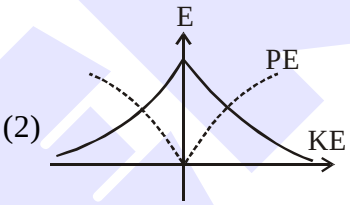
1. A particle moves with simple harmonic motion in a straight line. In first τ s, after starting from rest it travels a distance a, and in next τ s it travels 2a, in same direction, then : **[JEE Mains-2014]**

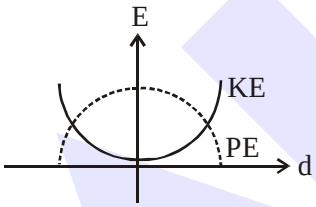
(1) Amplitude of motion is 4a
(2) Time period of oscillation is 6τ
(3) Amplitude of motion is 3a
(4) Time period of oscillation is 8τ

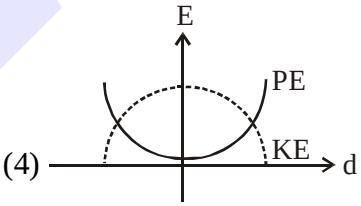
SH0119

2. For a simple pendulum, a graph is plotted between its kinetic energy (KE) and potential energy (PE) against its displacement d. Which one of the following represents these correctly ? (graphs are schematic and not drawn to scale) **[JEE Mains-2015]**

(1) 

(2) 

(3) 

(4) 

SH0120

3. A pendulum made of a uniform wire of cross sectional area A has time period T. When an additional mass M is added to its bob, the time period changes to T_M. If the Young's modulus of the material of the wire is Y then $\frac{1}{Y}$ is equal to :- (g = gravitational acceleration) **[JEE-Main-2015]**

(1) $\left[1-\left(\frac{T_M}{T}\right)^2\right]\frac{A}{Mg}$
(2) $\left[1-\left(\frac{T}{T_M}\right)^2\right]\frac{A}{Mg}$
(3) $\left[\left(\frac{T_M}{T}\right)^2-1\right]\frac{A}{Mg}$
(4) $\left[\left(\frac{T_M}{T}\right)^2-1\right]\frac{Mg}{A}$

EH0135

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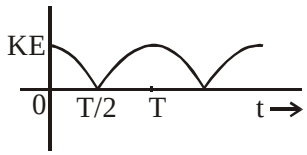
4. A particle performs simple harmonic motion with amplitude A. Its speed is trebled at the instant that it is at a distance $\frac{2A}{3}$ from equilibrium position. The new amplitude of the motion is :-

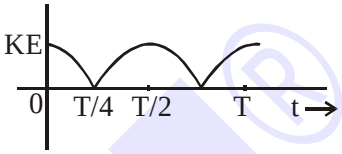
[JEE Mains-2016]

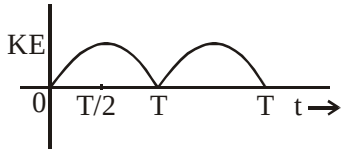
(1) $\frac{7A}{3}$ (2) $\frac{A}{3}\sqrt{41}$ (3) 3A (4) $A\sqrt{3}$

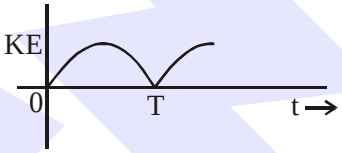
SH0121

5. A particle is executing simple harmonic motion with a time period T. AT time t = 0, it is at its position of equilibrium. The kinetic energy-time graph of the particle will look like **[JEE Main-2017]**

(1) 

(2) 

(3) 

(4) 

SH0122

6. A silver atom in a solid oscillates in simple harmonic motion in some direction with a frequency of 10^{12}/sec. What is the force constant of the bonds connecting one atom with the other ? (Mole wt. of silver =108 and Avagadro number = 6.02×10^{23} gm mole^{-1}) **[JEE Main-2018]**

(1) 7.1 N/m (2) 2.2 N/m (3) 5.5 N/m (4) 6.4 N/m

SH0123

SELECTED PROBLEMS FROM JEE-MAINS ONLINE PAPERS

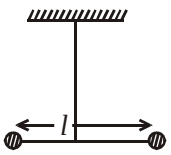
7. A rod of mass 'M' and length '2L' is suspended at its middle by a wire. It exhibits torsional oscillations; If two masses each of 'm' are attached at distance 'L/2' from its centre on both sides, it reduces the oscillation frequency by 20%. The value of ratio m/M is close to : **[JEE-Main-2019_Jan]**

(1) 0.17 (2) 0.37 (3) 0.57 (4) 0.77

SH0142

8. Two masses m and $\frac{m}{2}$ are connected at the two ends of a massless rigid rod of length *l*. The rod is suspended by a thin wire of torsional constant k at the centre of mass of the rod-mass system(see figure). Because of torsional constant k, the restoring torque is $\tau = k\theta$ for angular displacement θ. If the rod is rotated by θ_0 and released, the tension in it when it passes through its mean position will be:

[JEE-Main-2019_Jan]



(1) $\frac{3k\theta_0^2}{l}$ (2) $\frac{k\theta_0^2}{2l}$ (3) $\frac{2k\theta_0^2}{l}$ (4) $\frac{k\theta_0^2}{l}$

SH0143

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9. A particle executes simple harmonic motion with an amplitude of 5 cm. When the particle is at 4 cm from the mean position, the magnitude of its velocity in SI units is equal to that of its acceleration. Then, its periodic time in seconds is : **[JEE-Main-2019_Jan]**

(1) $\frac{7}{3}\pi$ (2) $\frac{3}{8}\pi$ (3) $\frac{4\pi}{3}$ (4) $\frac{8\pi}{3}$

SH0144

10. A simple pendulum of length 1 m is oscillating with an angular frequency 10 rad/s. The support of the pendulum starts oscillating up and down with a small angular frequency of 1 rad/s and an amplitude of 10^{-2} m. The relative change in the angular frequency of the pendulum is best given by :-

[JEE-Main-2019_Jan]

(1) 10^{-3} rad/s (2) 10^{-1} rad/s (3) 1 rad/s (4) 10^{-5} rad/s

SH0145

11. A pendulum is executing simple harmonic motion and its maximum kinetic energy is K_1. If the length of the pendulum is doubled and it performs simple harmonic motion with the same amplitude as in the first case, its maximum kinetic energy is K_2. Then :- **[JEE-Main-2019_Jan]**

(1) $K_2 = \frac{K_1}{4}$ (2) $K_2 = \frac{K_1}{2}$ (3) $K_2 = 2K_1$ (4) $K_2 = K_1$

SH0146

12. A simple harmonic motion is represented by:

$y = 5(\sin 3\pi t + \sqrt{3}\cos 3\pi t)$ cm

The amplitude and time period of the motion are: **[JEE-Main-2019_Jan]**

(1) 5cm, $\frac{3}{2}$s (2) 5cm, $\frac{2}{3}$s (3) 10cm, $\frac{3}{2}$s (4) 10cm, $\frac{2}{3}$s

SH0147

13. A simple pendulum oscillating in air has period T. The bob of the pendulum is completely immersed in a non-viscous liquid. The density of the liquid is $\frac{1}{16}$th of the material of the bob. If the bob is inside liquid all the time, its period of oscillation in this liquid is : **[JEE-Main-2019_April]**

(1) $4T\sqrt{\frac{1}{15}}$ (2) $2T\sqrt{\frac{1}{10}}$ (3) $4T\sqrt{\frac{1}{14}}$ (4) $2T\sqrt{\frac{1}{14}}$

SH0148

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14. A ring is hung on a nail. It can oscillate, without slipping or sliding (i) in its plane with a time period T_1 and, (ii) back and forth in a direction perpendicular to its plane, with a period T_2. the ratio $\frac{T_1}{T_2}$ will be : **[JEE-Main-2020_Sep]**

(1) $\frac{2}{\sqrt{3}}$ (2) $\frac{\sqrt{2}}{3}$ (3) $\frac{2}{3}$ (4) $\frac{3}{\sqrt{2}}$

SH0149

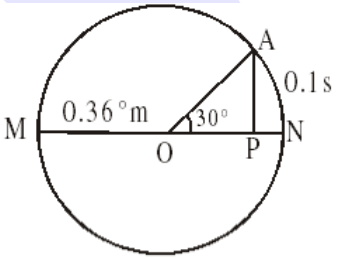
15. When a particle of mass m is attached to a vertical spring of spring constant k and released, its motion is described by $y(t) = y_0\sin^2\omega t$, where 'y' is measured from the lower end of unstretched spring. Then ω is : **[JEE-Main-2020_Sep]**

(1) $\sqrt{\frac{g}{y_0}}$ (2) $\sqrt{\frac{g}{2y_0}}$ (3) $\frac{1}{2}\sqrt{\frac{g}{y_0}}$ (4) $\sqrt{\frac{2g}{y_0}}$

SH0150

16. The point A moves with a uniform speed along the circumference of a circle of radius 0.36 m and covers 30° in 0.1 s. The perpendicular projection 'P' from 'A' on the diameter MN represents the simple harmonic motion of 'P'. The restoration force per unit mass when P touches M will be: **SH0259**

[JEE-Main-2021_Feb]



(1) 100 N (2) 0.49 N (3) 50 N (4) 9.87 N

17. The function of time representing a simple harmonic motion with a period of $\frac{\pi}{\omega}$ is : **SH0260**

[JEE-Main-2021_March]

(1) $\sin(\omega t) + \cos(\omega t)$ (2) $\cos(\omega t) + \cos(2\omega t) + \cos(3\omega t)$

(3) $\sin^2(\omega t)$ (4) $3\cos\left(\frac{\pi}{4} - 2\omega t\right)$

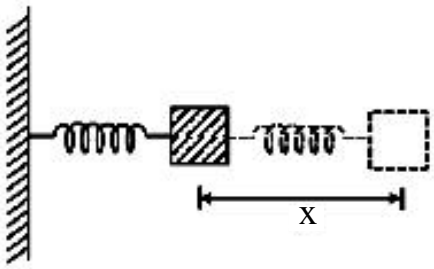
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18. A particle is making simple harmonic motion along the X-axis. If at a distances x_1 and x_2 from the mean position the velocities of the particle are v_1 and v_2 respectively. The time period of its oscillation is given as : **[JEE-Main-2021_July]**

(1) $T = 2\pi\sqrt{\frac{x_2^2 + x_1^2}{v_1^2 - v_2^2}}$ (2) $T = 2\pi\sqrt{\frac{x_2^2 + x_1^2}{v_1^2 + v_2^2}}$ (3) $T = 2\pi\sqrt{\frac{x_2^2 - x_1^2}{v_1^2 + v_2^2}}$ (4) $T = 2\pi\sqrt{\frac{x_2^2 - x_1^2}{v_1^2 - v_2^2}}$

SH0261

19. The motion of a mass on a spring, with spring constant K is as shown in figure. **SH0262**



The equation of motion is given by $x(t) = A\sin\omega t + B\cos\omega t$ with $\omega = \sqrt{\frac{K}{m}}$

Suppose that at time t = 0, the position of mass is x(0) and velocity v (0), then its displacement can also be represented as $x(t) = C\cos(\omega t - \phi)$, where C and ϕ are : **[JEE-Main-2021_July]**

(1) $C = \sqrt{\frac{2v(0)^2}{\omega^2} + x(0)^2}, \phi = \tan^{-1}\left(\frac{v(0)}{x(0)\omega}\right)$

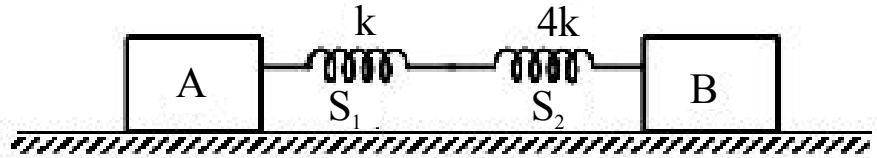
(2) $C = \sqrt{\frac{2v(0)^2}{\omega^2} + x(0)^2}, \phi = \tan^{-1}\left(\frac{x(0)\omega}{2v(0)}\right)$

(3) $C = \sqrt{\frac{v(0)^2}{\omega^2} + x(0)^2}, \phi = \tan^{-1}\left(\frac{x(0)\omega}{v(0)}\right)$

(4) $C = \sqrt{\frac{v(0)^2}{\omega^2} + x(0)^2}, \phi = \tan^{-1}\left(\frac{v(0)}{x(0)\omega}\right)$

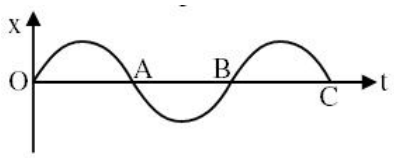
20. In the reported figure, two bodies A and B of masses 200 g and 800 g are attached with the system of springs. Springs are kept in a stretched position with some extension when the system is released. The horizontal surface is assumed to be frictionless. The angular frequency will be _____ rad/s when k = 20 N/m. **[JEE-Main-2021_July]**

SH0263

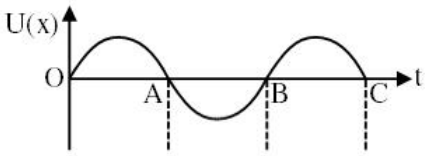


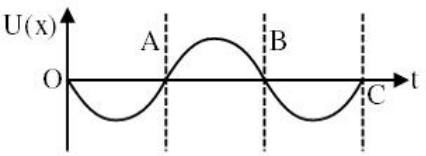
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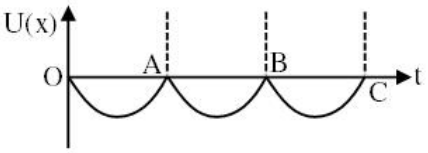
21. The variation of displacement with time of a particle executing free simple harmonic motion is shown in the figure. **[JEE-Main-2021_Aug]**

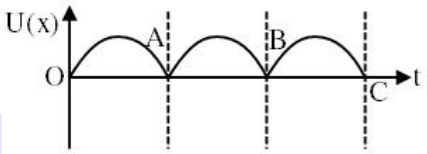


The potential energy U(x) versus time (t) plot of the particle is correctly shown in figure : **SH0264**

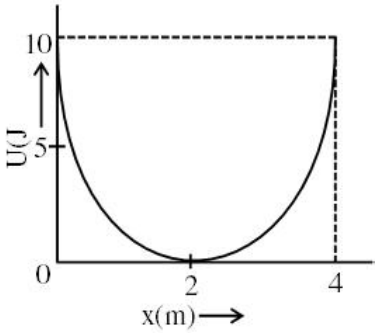
(1) 

(2) 

(3) 

(4) 

22. A mass of 5 kg is connected to a spring. The potential energy curve of the simple harmonic motion executed by the system is shown in the figure. A simple pendulum of length 4 m has the same period of oscillation as the spring system. What is the value of acceleration due to gravity on the planet where these experiments are performed? **[JEE-Main-2021_Aug]**



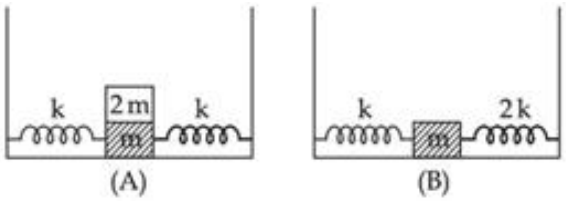
(1) 10 m/s^2

(2) 5 m/s^2

(3) 4 m/s^2

(4) 9.8 m/s^2 **SH0265**

23. In figure (A), mass '2 m' is fixed on mass 'm' which is attached to two springs of spring constant k. In figure (B), mass 'm' is attached to two spring of spring constant 'k' and '2k'. If mass 'm' in (A) and (B) are displaced by distance 'x' horizontally and then released, then time period T_1 and T_2 corresponding to (A) and (B) respectively follow the relation. **[JEE-Main-2022_July]**



(A) $\frac{T_1}{T_2}=\frac{3}{\sqrt{2}}$ (B) $\frac{T_1}{T_2}=\sqrt{\frac{3}{2}}$ (C) $\frac{T_1}{T_2}=\sqrt{\frac{2}{3}}$ (D) $\frac{T_1}{T_2}=\frac{\sqrt{2}}{3}$ **SH0266**

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24. A mass 0.9 kg, attached to a horizontal spring, executes SHM with an amplitude A_1. When this mass passes through its mean position, then a smaller mass of 124 g is placed over it and both masses move together with amplitude A_2. If the ratio $\frac{A_1}{A_2}$ is $\frac{\alpha}{\alpha-1}$, then the value of α will be ____. **SH0267**

[JEE-Main-2022_July]

25. The potential energy of a particle of mass 4 kg in motion along the x-axis is given by U = 4(1 – cos 4x) J. The time period of the particle for small oscillation $(\sin\theta \simeq \theta)$ is $\left(\frac{\pi}{K}\right)$ s. The value of K is …….. **[JEE-Main-2022_July]**
SH0268

26. The time period of oscillation of a simple pendulum of length L suspended from the roof of a vehicle, which moves without friction down an inclined plane of inclination α, is given by : **SH0269**
[JEE-Main-2022_July]

(A) $2\pi\sqrt{L/(g\cos\alpha)}$ (B) $2\pi\sqrt{L/(g\sin\alpha)}$

(C) $2\pi\sqrt{L/g}$ (D) $2\pi\sqrt{L/(g\tan\alpha)}$

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EXERCISE (JA)

1. A particle of mass m is attached to one end of a mass-less spring of force constant k, lying on a frictionless horizontal plane. The other end of the spring is fixed. The particle starts moving horizontally from its equilibrium position at time t = 0 with an initial velocity u_0. When the speed of the particle is 0.5 u_0, it collides elastically with a rigid wall. After this collision :- **[JEE-Advanced-2013]**

(A) the speed of the particle when it returns to its equilibrium position is u_0

(B) the time at which the particle passes through the equilibrium position for the first time is $t = \pi\sqrt{\frac{m}{k}}$

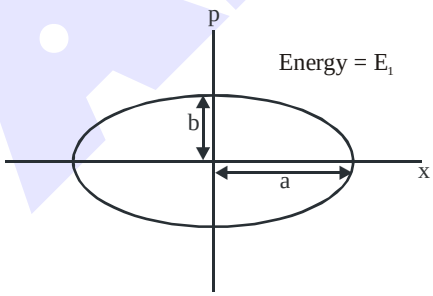
(C) the time at which the maximum compression of the spring occurs is $t = \frac{4\pi}{3}\sqrt{\frac{m}{k}}$

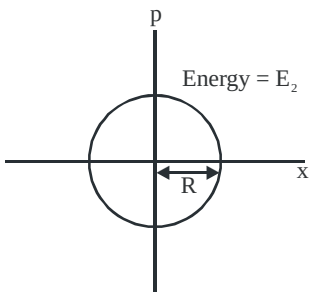
(D) the time at which the particle passes through the equilibrium position for the second time is $t = \frac{5\pi}{3}\sqrt{\frac{m}{k}}$

SH0129

2. Two independent harmonic oscillators of equal mass are oscillating about the origin with angular frequencies ω_1 and ω_2 and have total energies E_1 and E_2, respectively. The variations of their momenta p with positions x are shown in the figures. If $\frac{a}{b} = n^2$ and $\frac{a}{R} = n$, then the correct equation (s) is (are)

[JEE-Advanced-2015]





(A) $E_1\omega_1 = E_2\omega_2$ (B) $\frac{\omega_2}{\omega_1} = n^2$ (C) $\omega_1\omega_2 = n^2$ (D) $\frac{E_1}{\omega_1} = \frac{E_2}{\omega_2}$

SH0130

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3. A particle of unit mass is moving along the x-axis under the influence of a force and its total enegy is conserved. Four possible forms of the potential energy of the particle are given in column I (a and U_0 are constants). Match the potential energies in column I to the corresponding statement(s) in column-II **[JEE-Advanced-2015]**

Column-I		**Column-II**	
(A)	$U_1(x)=\frac{U_0}{2}\left[1-\left(\frac{x}{a}\right)^2\right]^2$	(P)	The force acting on the particle is zero at x = a.
(B)	$U_2(x)=\frac{U_0}{2}\left(\frac{x}{a}\right)^2$	(Q)	The force acting on the particle is zero at x = 0.
(C)	$U_3(x)=\frac{U_0}{2}\left(\frac{x}{a}\right)^2\exp\left[-\left(\frac{x}{a}\right)^2\right]$	(R)	The force acting on the particle is zero at x = –a.
(D)	$U_4(x)=\frac{U_0}{2}\left[\frac{x}{a}-\frac{1}{3}\left(\frac{x}{a}\right)^3\right]$	(S)	The particle experiences an attractive force towards x = 0 in the region |x| < a.
		(T)	The particle with total enegy $\frac{U_0}{4}$ can oscillate about the point x = –a.

SH0131

4. A block with mass M is connected by a massless spring with stiffness constant k to a rigid wall and moves without friction on a horizontal surface. The block oscillates with small amplitude A about an equilibrium position x_0. Consider two cases : (i) when the block is at x_0; and (ii) when the block is at $x = x_0 + A$. In both the cases, a particle with mass m (<M) is softly placed on the block after which they stick to each other. Which of the following statement(s) is(are) true about the motion after the mass m is placed on the mass M? **[JEE-Advanced-2016]**

(A) The amplitude of oscillation in the first case changes by a factor of $\sqrt{\frac{M}{m+M}}$, whereas in the second case it remains unchanged

(B) The final time period of oscillation in both the cases is same

(C) The total energy decreases in both the cases

(D) The instantaneous speed at x_0 of the combined masses decreases in both the cases.

SH0132

5. On a frictionless horizontal plane, a bob of mass m = 0.1 kg is attached to a spring with natural length ℓ_0 = 0.1 m. The spring constant is k_1 = 0.009 Nm^{-1} when the length of the spring $\ell > \ell_0$ and is k_2 = 0.016 Nm^{-1} when $\ell < \ell_0$. Initially the bob is released from ℓ = 0.15 m. Assume that Hooke's law remains valid throughout the motion. If the time period of the full oscillation is T = (nπ) s, then the integer closest to n is _____. **[JEE-Advanced-2022]**

SH0270

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ANSWER KEY

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	A	D	C	C	C	C	A	A	D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	B	B	D	B	A	A	A	B	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	A	A	C	B	D	C	C	B	A	C
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	C	C	C	A	B	C	D	A	D	B
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	C	A	A	A	B	D	D	A	C	A
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	B	D	C	B	C	B	C	B	A	C
Que.	61	62	63	64	65	66	67	68	69	70
Ans.	B	C	B	B	B	B	A	B	D	C
Que.	71									
Ans.	C									

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	D	A	B	A	C	A	B	A	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	C	A	C	A	B	A	B	A	A
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	D	B	C	A	A	C	B	B	C	B
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	C	B	A,D	B,C	B,C	A,B,C,D	A,D	A,B,D	A,C,D	A,C,D
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	A,B,C,D	A	C	C	A	C	B	D	D	C
Que.	51	52								
Ans.	B	D								

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EXERCISE # S

1. $y = 2\sin\left(\frac{10}{3}t - \frac{\pi}{2}\right)$ **2.** 0.33 sec **3.** 2 m/s **4.** $\frac{1}{2\pi}\sqrt{\frac{\beta}{\alpha}}$

5. (a) T/12, (b) T/8, (c) T/6, (d) T/4, (e) T/8 **6.** $x = (10\text{ cm})\sin\left[\left(\frac{\pi}{3}s^{-1}\right)t + \frac{\pi}{6}\right]$, $\frac{10}{9}\pi^2 \approx 11\text{ cm/s}^2$

7. 0.06m **8.** $y = 0.1\sin(4t + \pi/4)$ **9.** (i) $x_0 = 2$m; (ii) T = $\sqrt{2}\ \pi$ sec.; (iii) $2\sqrt{3}$

10. 3 **11.** $\omega = \sqrt{\frac{2T_0}{m\ell}}$ **12.** $1/\sqrt{3}$ **13.** M = 1.6 kg **14.** (a) $\frac{1}{T} = \frac{1}{2\pi}\sqrt{\frac{g}{L}}$, (b) $\frac{L}{3}$

15. 100 Nm^{-1} **16.** $10/\pi$ Hz, $\frac{5\sqrt{37}}{6}$ cm **17.** 2 **18.** 5 **19.** 1 **20.** $f = \frac{1}{2\pi}\sqrt{\frac{3g}{2R}}$

21. $\frac{5}{2\pi}$ Hz, 5 cm **22.** 0.16 kg **23.** $2\pi\sqrt{\frac{17L}{18g}}$ **24.** $\omega = \sqrt{\frac{k}{(m+6M)}}$

25. (a) 2 sec, (b) T = $\frac{2}{5^{1/4}}$ sec

26. (a) $T = mg + mgA^2\sin^2\sqrt{\frac{g}{\ell}}\,t$ (b) $t = (2n+1)\frac{\pi}{2}\sqrt{\frac{\ell}{g}}$; $n \in I$

$T_{max} = mg + mgA^2$

27. (b) $\frac{R}{\sqrt{2}}$ **28.** $2\pi\sqrt{\frac{r\sqrt{2}}{g}}$, $r/\sqrt{2}$ **29.** $-4\pi^2\cos\pi t$

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	4	3	1	2	1	2	4	4	1
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	4	1	1	2	4	4	4	4	10
Que.	21	22	23	24	25	26				
Ans.	4	3	A	16	2	A				

EXERCISE # JEE-ADVANCED

Que.	1	2	3	(A)	(B)	(C)	(D)	4	5	
Ans.	A,D	B,D		P,Q,R,T	Q,S	P,Q,R,S	P,R,T	A,B,D	6	

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